

Research report topic

Proposed title

Generating Rail Scenes Images for Semantic Segmentation Algorithms using Text-to-Image Diffusion Models with and without Conditional Control

Topic description

Usually, to obtain a good deep neural network model you need a lot of data for it to be trained on. In the context of rail scene semantic segmentation, there is not a lot of data to train such algorithms (the biggest dataset is RailSem19 containing 8500 pairs of image-masks, some of the annotations also not being that precise), so the motivation behind this research report would be to generate some data in the given scenario and to try to train a semantic segmentation algorithm to prove that some synthetic data would improve the performance of the current SOTA algorithms.

Bibliographic references

1. Zendel, Oliver, Markus Murschitz, Marcel Zeilinger, Daniel Steininger, Sara Abbasi and Csaba Beleznai. "RailSem19: A Dataset for Semantic Rail Scene Understanding." 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (2019): 1221-1229.
2. Zhang, Lvmin, Anyi Rao and Maneesh Agrawala. "Adding Conditional Control to Text-to-Image Diffusion Models." 2023 IEEE/CVF International Conference on Computer Vision (ICCV) (2023): 3813-3824.
3. Rombach, Robin, A. Blattmann, Dominik Lorenz, Patrick Esser and Björn Ommer. "High-Resolution Image Synthesis with Latent Diffusion Models." 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2021): 10674-10685.
4. Karras, Tero, Samuli Laine and Timo Aila. "A Style-Based Generator Architecture for Generative Adversarial Networks." 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2018): 4396-4405.
5. Le, Tuan Anh, Atilim Gunes Baydin, Robert Zinkov and Frank D. Wood. "Using synthetic data to train neural networks is model-based reasoning." 2017 International Joint Conference on Neural Networks (IJCNN) (2017): 3514-3521.

Chosen week for presentation

Week 11, at the course.